

Shor's Algorithm — Complete Cheat Sheet

modular periodicity $\xrightarrow{a^x \bmod N}$ cyclic shift $\xrightarrow{U_a \text{ on orbit}}$ characters $\xrightarrow{|u_s\rangle}$ roots of unity $\xrightarrow{e^{2\pi i s/r}}$ phase kickback $\xrightarrow{c-U}$ phase ladder $\xrightarrow{e^{2\pi i x s/r}}$ inverse QFT $\xrightarrow{\text{peak at } \ell \approx Qs/r}$ $\xrightarrow{\frac{\ell}{Q} \approx \frac{s}{r} \text{ cont. fractions}}$ $\rightarrow r \rightarrow \text{factors}$

Conventions (fix once)

$\mathbb{Z}/N\mathbb{Z}$, $(\mathbb{Z}/N\mathbb{Z})^\times$: integers / units mod N . $\omega_N = e^{2\pi i/N}$; i = imaginary unit (never an index).
QFT: $F_Q|x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} e^{2\pi i xy/Q} |y\rangle$, $F_Q^\dagger|x\rangle = \frac{1}{\sqrt{Q}} \sum_y e^{-2\pi i xy/Q} |y\rangle$.
Registers big-endian: $x = x_1 \cdots x_n$, $x = \sum_s x_s 2^{n-s}$, x_1 =MSB=top wire. Binary fraction $0.a_1 \cdots a_m = \sum_s a_s 2^{-s}$.
Gate: $c-U|c\rangle|\psi\rangle = |c\rangle U^c|\psi\rangle$. $R_k = \text{diag}(1, e^{2\pi i/2^k})$. Phases are mod 1.
 Sizes: $n = \lceil \log_2 N \rceil$, $Q = 2^t$ with $N^2 \leq Q < 2N^2$ (so $t = 2n + O(1)$).
Key sum: $\sum_{k=0}^{r-1} \omega_r^{k\ell} = r [\ell \equiv 0] \text{ (else 0)}$.

1. Problem & classical reduction

Goal: factor odd composite N (not a prime power). **Reduce to order finding**: pick a , $r = \text{ord}_N(a)$ = least r with $a^r \equiv 1$.
Square-root trick: if r even and $a^{r/2} \not\equiv -1 \pmod{N}$ then $x = a^{r/2}$ is a non-trivial $\sqrt{1}$ ($x^2 \equiv 1$, $x \not\equiv \pm 1$), so
 $\gcd(a^{r/2} - 1, N)$, $\gcd(a^{r/2} + 1, N)$ are proper factors.
 (From $a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1)$.)
Random a works w.p. $\geq 1 - 2^{1-m} \geq \frac{1}{2}$ (m =#distinct primes).
Whole algorithm:
 1. if N even / prime power: output factor classically.
 2. random $a \in \{2, \dots, N-1\}$; if $\gcd(a, N) > 1$ done (lucky).
 3. **[quantum]** order-find $\rightarrow \ell$ (Box 8).
 4. continued fractions on $\ell/Q \rightarrow s'/r'$ (Box 11).
 5. verify $a^{r'} \equiv 1$; else test small multiples / repeat.
 6. extract factors via the gcds. Verifies itself (Las Vegas).

2. Minimal quantum mechanics

State $|\psi\rangle = \sum_x \alpha_x |x\rangle$, $\sum |\alpha_x|^2 = 1$; $\alpha_x \in \mathbb{C}$ is an *amplitude*, **not** a probability (amplitudes can cancel).
 n qubits: $\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$. Tensor $\otimes$; a state that is not a single $|\alpha\rangle \otimes |\beta\rangle$ is *entangled*.
Unitary U : $U^\dagger U = \mathbb{I}$, reversible. **Controlled $c-U$** $|c\rangle|\psi\rangle = |c\rangle U^c|\psi\rangle$.
Born rule: measuring $\sum_x \alpha_x |x\rangle |\chi_x\rangle$ gives x w.p. $|\alpha_x|^2$.
Phase: global $e^{i\gamma}|\psi\rangle$ unobservable; *relative* phase is physical. $|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}$: same Pr in Z , but $H|\pm\rangle = |0/1\rangle$. $e^{i\theta}$ = rotation by θ .

3. Interference (name the terms)

Under unitary V : $\beta_y = \sum_x V_{yx} \alpha_x$, $\Pr[y] = |\beta_y|^2$. Amplitudes add *before* squaring:
 $|a+b|^2 = |a|^2 + |b|^2 + 2\text{Re}(a\bar{b}) = r^2 + s^2 + 2rs \cos(\theta - \varphi)$.
 Only contributions to the *same* $|y\rangle$ add; a label in an orthogonal unmeasured register kills cross terms ($\langle \chi_x | \chi_{x'} \rangle = 0$).

4. The hidden symmetry (why it works)

Modular mult: $U_a|y\rangle = |ay \bmod N\rangle$ for $y < N$, $= |y\rangle$ else (reversible; $U_a^p = U_{a^p \bmod N}$).
 On the orbit $\mathcal{O} = \{a^k \bmod N\}$ ($|\mathcal{O}| = r$), $|k\rangle \leftrightarrow |a^k\rangle$ makes U_a the **cyclic shift** S .

Eigenvectors = characters:

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^k \bmod N\rangle, \quad U_a|u_s\rangle = e^{2\pi i s/r} |u_s\rangle.$$

Eigenvalue eqn forces $\lambda^r = 1 \Rightarrow \lambda = \omega_r^s$; period r = common denominator of eigenphases s/r .

Free input: $|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle$ (equal mix of all eigenvectors – no need to know r).

Fourier = **change of basis** to eigenbasis: $S = F D F^\dagger$, $D = \text{diag}(\omega_r^s)$. Spectral theorem: simple spectrum $\Rightarrow$ this basis is *unique*, not a trick.

5. Phase kickback (derived)

If $U|u\rangle = e^{2\pi i \varphi} |u\rangle$ then

$$c-U[(\alpha|0\rangle + \beta|1\rangle)|u\rangle] = (\alpha|0\rangle + \beta e^{2\pi i \varphi} |1\rangle)|u\rangle.$$

Eigenvalue $\rightarrow$ *relative* phase on control; target unchanged (a catalyst). With $|+\rangle$ then H : $\Pr[0] = \cos^2 \pi \varphi$, $\Pr[1] = \sin^2 \pi \varphi$.
 $c-U$ does **not** force the target into an eigenstate; a superposition target $\Rightarrow$ entanglement (branches tagged by orthogonal $|u_s\rangle$).

6. Phase estimation / phase ladder

t counting qubits, $Q = 2^t$. $H^{\otimes t} \rightarrow \frac{1}{\sqrt{Q}} \sum_x |x\rangle |u\rangle$. Apply $c-U^{2^{t-s}}$ (qubit s), using $U^{2^j} |u\rangle = e^{2\pi i 2^j \varphi} |u\rangle$:

$$\bigotimes_{s=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-s} \varphi} |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x \varphi} |x\rangle$$

(uses $x = \sum_s x_s 2^{t-s}$). Before iQFT all $|\text{amp}\rangle = 1/\sqrt{Q}$: *uniform*; info is in relative phases. U^{2^j} built by j *squarings* (cheap) – source of speed-up.
 On $|1\rangle = \frac{1}{\sqrt{r}} \sum_s |u_s\rangle$: each branch s carries ramp $e^{2\pi i x s/r}$; measuring samples one s (random). Same state, two readings:

$$\frac{1}{\sqrt{Q}} \sum_x |x\rangle |a^x\rangle = \frac{1}{\sqrt{r}} \sum_s \left(\frac{1}{\sqrt{Q}} \sum_x e^{2\pi i x s/r} |x\rangle \right) |u_s\rangle.$$

7. Inverse QFT = coherent unwinding

On one ramp, $F_Q^\dagger$ gives amplitude at y

$$A(y) = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x \delta_y}, \quad \delta_y = \varphi - \frac{y}{Q}.$$

$\delta_y=0 \Rightarrow A=1$; $\delta_y=k/Q$ ($k \neq 0$) $\Rightarrow A=0$ (roots-of-unity polygon); near 0 $\Rightarrow$ arc, large. Closed form

$$\Pr[y] = |A(y)|^2 = \frac{1}{Q^2} \frac{\sin^2(\pi Q \delta_y)}{\sin^2(\pi \delta_y)} \text{ (Fejér)}.$$

Nearest grid point gets $\geq 4/\pi^2 \approx 0.405$. Mixture over branches: $\Pr[y] = \frac{1}{r} \sum_s |A_s(y)|^2$ (no cross terms). Peak at $\ell \approx Qs/r$.

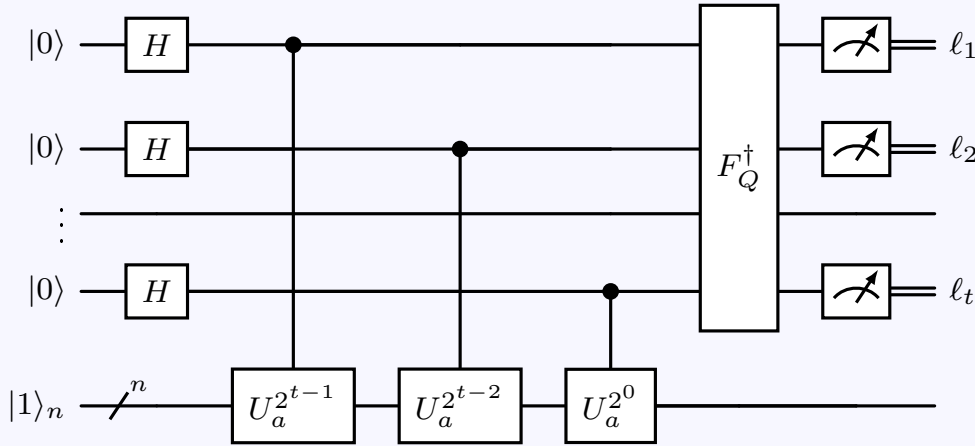
11. Continued fractions $\rightarrow r$

Measured ℓ : $\ell/Q \approx s/r$. **Legendre:** $|x - \frac{p}{q}| < \frac{1}{2q^2} \Rightarrow \frac{p}{q}$ is a convergent of x . Since $Q \geq N^2$, error $< \frac{1}{2r^2}$, so s/r is recoverable.

Procedure: expand ℓ/Q ; take convergents p_k/q_k with $q_k < N$; the denominator $r' = r/\gcd(s, r)$ is a *divisor* of r . Verify $a^{r'} \equiv 1$; if only a divisor, test $2r', 3r', \dots$ or combine runs by lcm. Direct success w.p. $\varphi(r)/r$.

8. Full order-finding circuit

Phase estimation with $U = U_a$, work register in the free state $|1\rangle_n$. Each $c\text{-}U_a^{2^{t-s}}$ is one modular mult by a classical constant (Box 10).



Top wire (MSB) drives $U_a^{2^{t-1}}$. Whole ladder = modular exponentiation $|x\rangle|1\rangle \mapsto |x\rangle|a^x \bmod N\rangle$. Work register returns to $|1\rangle$, never measured. Read $\ell = \ell_1 \dots \ell_t$, feed to continued fractions.

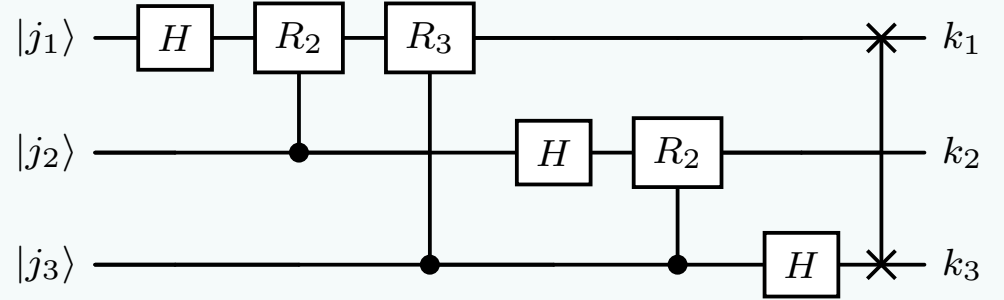
9. Building the QFT and inverse QFT

Product form ($Q = 2^n$, MSB first):

$$F_{2^n} |j_1 \dots j_n\rangle = \frac{1}{2^{n/2}} \bigotimes_{\ell=1}^n (|0\rangle + e^{2\pi i 0.j_{n-\ell+1} \dots j_n} |1\rangle).$$

So each output qubit is a $|0\rangle + e^{2\pi i \phi}|1\rangle$: H writes the leading bit, each controlled- R_k appends bit k . $R_k = \text{diag}(1, e^{2\pi i/2^k})$ ($R_1=Z$).

Circuit ($n=3$):



Recipe: for $s = 1..n$: H on qubit s , then $c\text{-}R_k$ (control $s+k-1$) for $k = 2, \dots, n-s+1$; finish with $\lfloor n/2 \rfloor$ SWAPS (the circuit outputs qubits reversed). Count: $n H + \binom{n}{2}$ rotations + swaps = $\Theta(n^2)$.

Inverse QFT $F_Q^\dagger$: reverse the gate order and conjugate every rotation, $R_k \rightarrow R_k^\dagger = \text{diag}(1, e^{-2\pi i/2^k})$ (H, SWAP self-inverse). This is the box $F_Q^\dagger$ in Box 8.

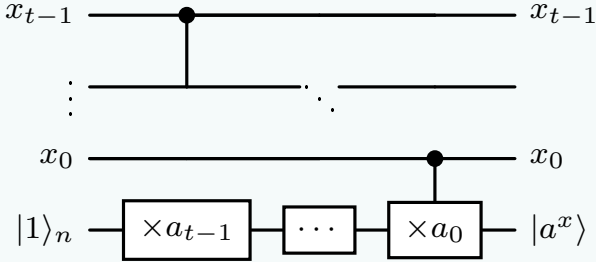
Approximate QFT: drop R_k with $k > m = O(\log(n/\epsilon))$; error $\leq 2\pi n/2^m$, cost $O(n \log n)$ – the version actually used.

10. Building the c - U gates

The ladder needs c - $U_a^{2^j}$. **Key:** $U_a^{2^j} = U_{a_j}$ with $a_j := a^{2^j} \bmod N$ precomputed *classically* ($a_0=a$, $a_{j+1}=a_j^2 \bmod N$). So each c - U^{2^j} is **one controlled modular multiplication by a classical constant**. Reduce it downward:

$$\underbrace{\text{mod. exp.}}_{|x\rangle|1\rangle \rightarrow |x\rangle|a^x\rangle} \Leftarrow \underbrace{\text{mod. mult}}_{|x\rangle \rightarrow |ax \bmod N\rangle} \Leftarrow \underbrace{\text{mod. add}}_{|x\rangle \rightarrow |x+c \bmod N\rangle} \Leftarrow \underbrace{\text{integer add}}_{|a\rangle|b\rangle \rightarrow |a\rangle|a+b\rangle}.$$

(i) **Modular exponentiation** = ladder of controlled modular multipliers ($2n$ mults total):



(ii) **Modular mult by constant a** (out-of-place): $|x\rangle|0\rangle \rightarrow |x\rangle|ax \bmod N\rangle = \prod_{i=0}^{n-1} \text{ADD}_{\bmod N}(2^i a \bmod N)$ controlled on $x_i - n$ controlled modular adds, $\Theta(n^2)$ Toffoli. *In-place* U_a : out-of-place mult, n SWAPS, then inverse mult by $a^{-1} \bmod N$ (needs $\gcd(a, N)=1$).

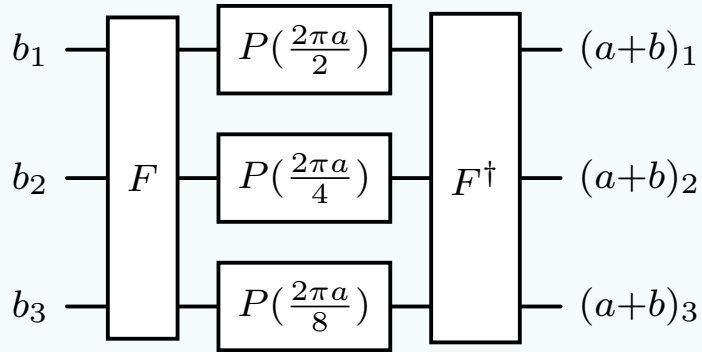
(iii) **Modular add by constant c** (Th. 10.1, 1 ancilla, $\Theta(n)$ Toffoli): $\text{ADD}(c) \rightarrow \text{SUB}(N) \rightarrow [\text{sign} \rightarrow t] \rightarrow \text{ADD}(N)^t \rightarrow \text{SUB}(c) \rightarrow [\text{reset } t] \rightarrow \text{ADD}(c)$; ≈ 5 constant integer additions.

10'. Integer adder (bottom of the stack)

Ripple-carry (CDKM): MAJ/UMA chain, $|a\rangle|b\rangle \rightarrow |a\rangle|a+b\rangle$, $2n$ Toffoli, $2n+2$ qubits, *no garbage* (compute-copy-uncompute). Temporary-AND (Gidney) $\Rightarrow 4T$ per Toffoli pair.

Draper (Fourier) adder, no ancilla – ties back to Box 9:

$$\text{ADD}(a) = F^\dagger \left[\bigotimes_{j=1}^n P\left(\frac{2\pi a}{2^j}\right) \right] F, \quad P(\theta) = \text{diag}(1, e^{i\theta}).$$



For a *quantum* addend, replace each P by controlled rotations; adjacent Draper adders share $F/F^\dagger$ (they cancel) – gives Beauregard's $2n+3$ -qubit Shor circuit. Ripple-carry wins on Toffoli count fault-tolerantly.

12. Worked examples

Dyadic, $N = 15, a = 2$: orbit $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 1$, $r = 4$ ($4 \mid Q$, exact). $Q = 8$: outcomes $\ell \in \{0, 2, 4, 6\}$ each $\frac{1}{4}$. $\ell=2 \rightarrow 1/4 \rightarrow r=4$; $\ell=4 \rightarrow 1/2 \rightarrow r'=2$ (divisor, test $\times 2$). $\gcd(2^2 \mp 1, 15) = 3, 5$.

Non-dyadic, $N = 21, a = 2$: orbit $1, 2, 4, 8, 16, 11$, $r = 6$ ($6 \nmid Q$, peaks *broaden*). $Q = 512$: e.g. $s=1 \Rightarrow \ell \approx 85$, CF $[0; 6, 42, 2] \rightarrow 1/6 \rightarrow r=6$. $s=1, 5$ ($\gcd=1$) succeed directly; $s=2, 3, 4$ give divisors $3, 2, 3$. $\gcd(2^3 \mp 1, 21) = 7, 3$.

13. Resources & pitfalls

Qubits $\Theta(n)$ (e.g. $2n+3$). **Gates** dominated by modular exponentiation: $2n$ mults $\times n$ adds $\times \Theta(n)$ each = $\Theta(n^3)$ Toffoli ($\approx 0.3n^3$ best); QFT is only $\Theta(n^2) / O(n \log n)$ – *not* the bottleneck. Success: $\geq 4/\pi^2$ per peak, $\varphi(r)/r$ per run; $O(\log \log N)$ repeats.

Don't: treat amplitudes as probabilities; say “tries all answers”; claim c - U forces an eigenstate; attribute speed-up to QFT gate count; say “phases interfere” without naming terms; forget the closing SWAPS / the $R_k \rightarrow R_k^\dagger$ conjugation in $F^\dagger$.